

Gradient (Gradcal) Calibration Setup and Overview

([Click here for Summary Version](#))

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Timing Not Applicable	1 hrs	Timing Not Applicable

2 Overview

In this procedure, the strength of the gradient fields are calibrated by modifying the control variables cfxfull, cfyfull, and cfzfull. A separate scan will be taken to calibrate each of the three gradient fields (x, y, and z).

NOTE: For TwinSpeed, the whole procedure must be repeated for each gradient mode (GradMode) separately, Whole-Body (WB) and Zoom (ZM) gradients.

NOTE: *For Mobile MR Systems, system-specific procedure. This procedure is to be performed after a Mobile MR System move, at the beginning of the day, or any time shim is suspected of affecting image quality.*

NOTE: This procedure must be performed prior to Grafidy and Main Shimming calibrations.

Last Update	09/30/2004
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3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
DQA Phantom	1	-	2371512	-

3.2 Consumables

No consumables required.

3.3 Replacement Parts

No replacement parts required.

3.4 Safety



POISON HAZARD!

1.0T AND 1.5T PHANTOMS CONTAINS NICKEL, A SUSPECT CARCINOGEN. DO NOT INGEST.



DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND FEDERAL REGULATIONS.



⚠ WARNING

POISON HAZARD!

3.0T PHANTOMS CONTAINS SILICON OIL, DO NOT INGEST.

DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND FEDERAL REGULATIONS.

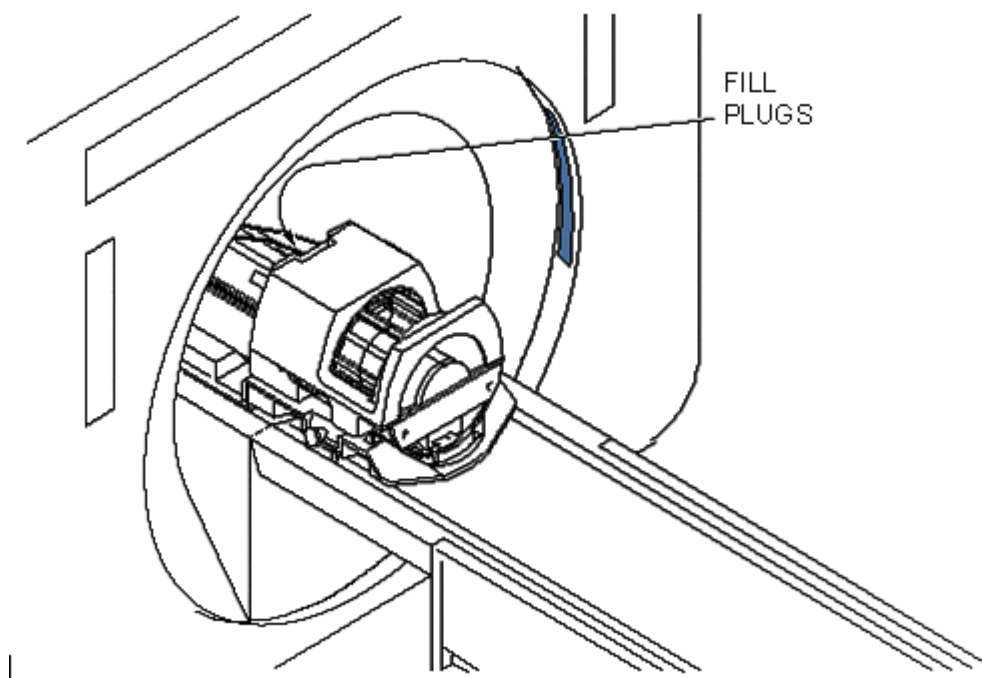
3.5 Required Conditions

No required conditions.

4 Procedure

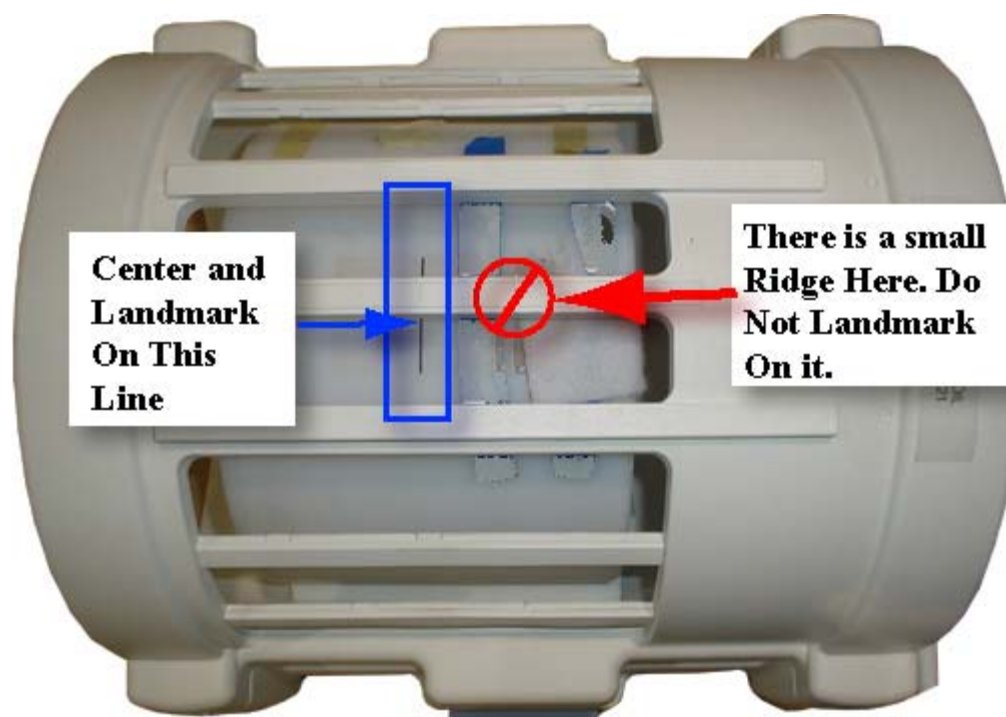
1. Place the DQA phantom in the head coil. Position it with the fill plugs up, and toward the rear of the magnet (see [Illustration 1](#)).

Illustration 1: DQA PHANTOM POSITIONING



2. Landmark on the phantom in the sagittal and axial planes (DQA coronal plane is not at isocenter). See [Illustration 2](#) for positioning the phantom in the quad head coil.

Illustration 2: LANDMARKING DQA PHANTOM



3. At the keypad on the front magnet enclosure, press <LANDMARK> then <MOVE TO SCAN>.

NOTE: Do not rotate or skew the phantom in the head coil. This will cause errors in the analysis of Gradient Calibration. Landmark errors can also cause errors in the analysis.

4. Click on [Autoview], just below the Autoview image display screen. Your scanned images will be displayed automatically
5. All three plans must be scanned and analyzed in order to complete this process. To Start the process you must first start the Gradcal tool. See [Gradient Calibration Image Analysis](#)

5 Finalization

1. Perform a test scan to insure the system is running properly.
2. Remove and put away phantoms used during this procedure.

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